IoT edge applications," says Matt Johnson, CEO of Silicon Labs. These SoCs will be available for purchase in the second half of 2022.

MAX78000 Development Board by Maxim Integrated: The MAX78000 is an AI microcontroller built to enable neural networks to execute at ultra-low power. It has a hardware based convolutional neural network (CNN) accelerator, which enables battery-powered applications to execute AI inferences.

Its CNN engine has a weight storage memory of 442kB and can support 1-, 2-, 4-, and 8-bit weights. Being SRAM based, the CNN memory enables AI network updates to happen on the fly. The CNN architecture is very flexible, allowing networks to be trained in conventional toolsets like PyTorch and TensorFlow.

Kria KV260 Vision AI Starter Kit by Xilinx: This is a development platform for Xilinx's K26 system on module (SOM), which specifically targets vision AI applications in smart cities and smart factories. These SOMs are tailored to enable rapid deployment in edge based applications.

Because it is based on an FPGA, the programmable logic allows users to implement custom accelerators for vision and ML functions. "With Kria, our initial focus was Vision AI in smart cities and, to some extent, in medical applications. One of the things that we are focused on now and moving forward is expanding into robotics and other factory applications," says Chetan Khona, director of Industrial, Vision, Healthcare & Sciences markets at Xilinx.

Gluon AI co-processor by AlphaICs: The Gluon AI accelerator is optimised for vision applications and provides maximum throughput with minimum latency and low power. It comes with an SDK that ensures easy deployment of neural networks.

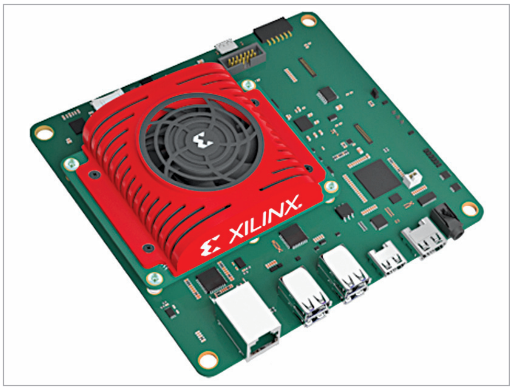

Figure 8: Kria KV260 Vision AI Starter Kit (Credit: Xilinx)

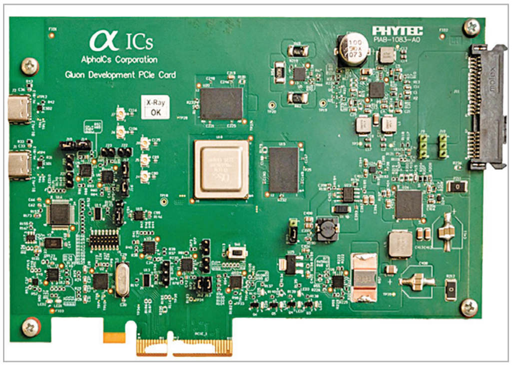

Figure 9: Gluon Evaluation Board (Credit: AlphaICs)

Figure 10: Intel NCS2 (Credit: Intel)

AlphaICs is currently sampling this accelerator for early customers. It is engineered for OEMs and solution providers targeting vision market segments, such as surveillance, industrial, retail, industrial IoT, and edge gateway manufacturers. The company also offers an evaluation board that can be used to prototype and develop AI hardware.

Intel's Neural Compute Stick 2 (Intel NCS2): Intel's NCS2 looks like a USB pen drive but actually brings AI and computer vision to edge devices in a very easy manner. It contains a dedicated hardware accelerator for deep neural network interfaces and is built on the Movidius Myriad X Vision processing unit (VPU).

Apart from using the NCS2 for their PC applications, designers can even use it with edge devices like Raspberry Pi 3. This will make prototyping very easy and will enhance applications like drones, smart cameras, and robots. Setting up and configuring its software environment is not complex, and detailed tutorials and instructions on various projects are provided by Intel.

A long way to go

The field of AI accelerators is still niche, and there is still a lot of room for innovation. "Many great ideas were implemented in the past five years, but even those are a fraction of the staggering number of AI accelerator designs and ideas in academic papers. There are still many ideas that can trickle-down from the academic world to industry practitioners," says Fuchs. END

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